

Question	Scheme	Marks	AOs
14(a)	$\frac{dV}{dt} = 0.48 - 0.1h$	B1	3.1b
	$V = 24h \Rightarrow \frac{dV}{dh} = 24 \text{ or } \frac{dh}{dV} = \frac{1}{24}$	B1	3.1b
	$\frac{dh}{dt} = \frac{dV}{dt} \times \frac{dh}{dV} = \frac{0.48 - 0.1h}{24}$ or e.g. $\frac{dV}{dt} = \frac{dV}{dh} \frac{dh}{dt} \Rightarrow 0.48 - 0.1h = 24 \frac{dh}{dt}$	M1	2.1
	$1200 \frac{dh}{dt} = 24 - 5h^*$	A1*	1.1b
		(4)	
(b)	$1200 \frac{dh}{dt} = 24 - 5h \Rightarrow \int \frac{1200}{24 - 5h} dh = \int dt$ $\Rightarrow \text{e.g. } \alpha \ln(24 - 5h) = t(+c) \text{ oe}$ or $1200 \frac{dh}{dt} = 24 - 5h \Rightarrow \frac{dt}{dh} = \frac{1200}{24 - 5h}$ $\Rightarrow \text{e.g. } t(+c) = \alpha \ln(24 - 5h) \text{ oe}$	M1	3.1a
	$t = -240 \ln(24 - 5h)(+c) \text{ oe}$	A1	1.1b
	$t = 0, h = 2 \Rightarrow 0 = -240 \ln(24 - 10) + c \Rightarrow c = \dots(240 \ln 14)$	M1	3.4
	$t = 240 \ln(14) - 240 \ln(24 - 5h)$	A1	1.1b
	$t = 240 \ln \frac{14}{24 - 5h} \Rightarrow \frac{t}{240} = \ln \frac{14}{24 - 5h} \Rightarrow e^{\frac{t}{240}} = \frac{14}{24 - 5h}$ $\Rightarrow 14e^{-\frac{t}{240}} = 24 - 5h \Rightarrow h = \dots$	ddM1	2.1
	$h = 4.8 - 2.8e^{-\frac{t}{240}} \text{ oe e.g. } h = \frac{24}{5} - \frac{14}{5}e^{-\frac{t}{240}}$	A1	3.3
		(6)	
(c)	Examples: - As $t \rightarrow \infty, e^{-\frac{t}{240}} \rightarrow 0$ - When $h > 4.8, \frac{dV}{dt} < 0$ - Flow in = flow out at max h so $0.1h = 4.8 \rightarrow h = 4.8$ - As $e^{-\frac{t}{240}} > 0, h < 4.8$ $h = 5 \Rightarrow \frac{dV}{dt} = -0.02 \text{ or } \frac{dh}{dt} = -\frac{1}{1200}$ $\frac{dh}{dt} = 0 \Rightarrow h = 4.8$ $h = 5 \Rightarrow 4.8 - 2.8e^{-\frac{t}{240}} = 5 \Rightarrow e^{-\frac{t}{240}} < 0$	M1	3.1b
	- The limit for h (according to the model) is 4.8m and the tank is 5m high so the tank will never become full - If $h = 5$ the tank would be emptying so can never be full - The equation can't be solved when $h = 5$	A1	3.2a
		(2)	

Question	Scheme	Marks	AOs
10(a)	$\frac{dH}{dt} = \frac{H \cos 0.25t}{40} \Rightarrow \int \frac{1}{H} dH = \int \frac{\cos 0.25t}{40} dt$	M1	3.1a
	$\ln H = \frac{1}{10} \sin 0.25t (+c)$	M1 A1	1.1b 1.1b
	Substitutes $t = 0, H = 5 \Rightarrow c = \ln(5)$	dM1	3.4
	$\ln\left(\frac{H}{5}\right) = \frac{1}{10} \sin 0.25t \Rightarrow H = 5e^{0.1 \sin 0.25t} *$	A1*	2.1
		(5)	
(b)	Max height = $5e^{0.1} = 5.53 \text{ m}$ (Condone lack of units)	B1	3.4
		(1)	
(c)	Sets $0.25t = \frac{5\pi}{2}$	M1	3.1b
	31.4	A1	1.1b
		(2)	

(8 marks)**(a)**

M1: Separates the variables to reach $\int \frac{1}{H} dH = \int \frac{\cos 0.25t}{40} dt$ or equivalent.

The integral signs need to be present on both sides and the dH AND dt need to be in the correct positions.

M1: Integrates both sides to reach $\ln H = A \sin 0.25t$ or equivalent with or without the $+c$

A1: $\ln H = \frac{1}{10} \sin 0.25t + c$ or equivalent with or without the $+c$. Allow two constants, one either side

If the 40 was on the lhs look for $40 \ln H = 4 \sin 0.25t + c$ or equivalent.

dM1: Substitutes $t = 0, H = 5 \Rightarrow c = ..$ There needs to have been a single " $+c$ " to find.

It is dependent upon the previous M mark. You may allow even if you don't explicitly see " $t = 0, H = 5$ " as it may be implied from their previous line

If the candidate has attempted to change the subject and made an error/ slip then condone it for this M but not the final A. Eg. $40 \ln H = 4 \sin 0.25t + c \Rightarrow H^{40} = e^{4 \sin 0.25t} + e^c \Rightarrow 5^{40} = 1 + e^c \Rightarrow c = ..$

Also many students will be attempting to get to the given answer so condone the method of finding $c = ..$
These students will lose the A1* mark

A1*: Proceeds via $\ln H = \frac{1}{10} \sin 0.25t + \ln 5$ or equivalent to the given answer $H = 5e^{0.1 \sin 0.25t}$ with at least one correct intermediate line and no incorrect work.

DO NOT condone c 's going to c 's when they should be e^c or A

Question	Scheme	Marks	AOs
10 (a)	$\frac{dr}{dt} \propto \pm \frac{1}{r^2}$ or $\frac{dr}{dt} = \pm \frac{k}{r^2}$ (for k or a numerical k)	M1	3.3
	$\int r^2 dr = \int \pm k dt \Rightarrow \dots$ (for k or a numerical k)	M1	2.1
	$\frac{1}{3} r^3 = \pm kt \{+ c\}$	A1	1.1b
	$t=0, r=5$ and $t=4, r=3$ gives $\frac{1}{3} r^3 = -\frac{49}{6}t + \frac{125}{3}$, where r, in mm, is the radius {of the mint} and t, in minutes, is the time from when it {the mint} was placed in the mouth $t=0, r=5$ and $t=240, r=3$ gives $\frac{1}{3} r^3 = -\frac{49}{360}t + \frac{125}{3}$, where r, in mm, is the radius {of the mint} and t, in seconds, is the time from when it {the mint} was placed in the mouth	M1	3.1a
		A1	1.1b
		(5)	
(b)	$r=0 \Rightarrow 0 = -\frac{49}{6}t + \frac{125}{3} \Rightarrow 0 = -49t + 250 \Rightarrow t = \dots$	M1	3.4
	time = 5 minutes 6 seconds	A1	1.1b
		(2)	
(c)	Suggests a suitable limitation of the model. E.g. • Model does not consider how the mint is sucked • Model does not consider whether the mint is bitten • Model is limited for times up to 5 minutes 6 seconds, o.e. • Not valid for times greater than 5 minutes 6 seconds, o.e. • Mint may not retain the shape of a sphere (or have uniform radius) as it is being sucked • The model indicates that the radius of the mint is negative after it dissolves • Model does not consider the temperature in the mouth • Model does not consider rate of saliva production • Mint could be swallowed before it dissolves in the mouth	B1	3.5b
		(1)	

(8 marks)

Question	Scheme	Marks	AOs
14	$N = \frac{900}{3 + 7e^{-0.25t}} = 900(3 + 7e^{-0.25t})^{-1}, t \in \mathbb{R}, t \geq 0; \quad \frac{dN}{dt} = \frac{N(300 - N)}{1200}$		
(a)	90	B1	3.4
		(1)	
(b) Way 1	$\frac{dN}{dt} = -900(3 + 7e^{-0.25t})^{-2} (7(-0.25)e^{-0.25t}) \quad \left\{ = \frac{900(0.25)(7)e^{-0.25t}}{(3 + 7e^{-0.25t})^2} \right\}$	M1	2.1
		A1	1.1b
	$\Rightarrow \frac{dN}{dt} = \frac{900(0.25) \left(\left(\frac{900}{N} - 3 \right) \right)}{\left(\frac{900}{N} \right)^2}$	dM1	2.1
	correct algebra leading to $\frac{dN}{dt} = \frac{N(300 - N)}{1200} \quad *$	A1*	1.1b
		(4)	
(b) Way 2	$\frac{dN}{dt} = -900(3 + 7e^{-0.25t})^{-2} (7(-0.25)e^{-0.25t}) \quad \left\{ = \frac{900(0.25)(7)e^{-0.25t}}{(3 + 7e^{-0.25t})^2} \right\}$	M1	2.1
		A1	1.1b
	$\frac{N(300 - N)}{1200} = \frac{\left(\frac{900}{3 + 7e^{-0.25t}} \right) \left(300 - \frac{900}{3 + 7e^{-0.25t}} \right)}{1200}$	dM1	2.1
	LHS = $\frac{1575e^{-0.25t}}{(3 + 7e^{-0.25t})^2}$ o.e., RHS = $\frac{900(300(3 + 7e^{-0.25t}) - 900)}{1200(3 + 7e^{-0.25t})^2} = \frac{1575e^{-0.25t}}{(3 + 7e^{-0.25t})^2}$ o.e. and states hence $\frac{dN}{dt} = \frac{N(300 - N)}{1200}$ (or LHS = RHS) *	A1*	1.1b
		(4)	
(c)	Deduces $N = 150$ (can be implied)	B1	2.2a
	so $150 = \frac{900}{3 + 7e^{-0.25T}} \Rightarrow e^{-0.25T} = \frac{3}{7}$	M1	3.4
	$T = -4 \ln\left(\frac{3}{7}\right)$ or $T = \text{awrt } 3.4$ (months)	dM1	1.1b
		A1	1.1b
		(4)	
(d)	either one of 299 or 300	B1	3.4
		(1)	
(10 marks)			

Question	Scheme	Marks	AOs
14 (a)	$\{u = 4 - \sqrt{h} \Rightarrow\} \frac{du}{dh} = -\frac{1}{2}h^{-\frac{1}{2}} \text{ or } \frac{dh}{du} = -2(4-u) \text{ or } \frac{dh}{du} = -2\sqrt{h}$	B1	1.1b
	$\left\{ \int \frac{dh}{4-\sqrt{h}} = \right\} \int \frac{-2(4-u)}{u} du$	M1	2.1
	$= \int \left(-\frac{8}{u} + 2 \right) du$	M1	1.1b
	$= -8\ln u + 2u \{+c\}$	M1	1.1b
		A1	1.1b
	$= -8\ln 4-\sqrt{h} + 2(4-\sqrt{h}) + c = -8\ln 4-\sqrt{h} - 2\sqrt{h} + k *$	A1*	2.1
		(6)	
(b)	$\left\{ \frac{dh}{dt} = \frac{t^{0.25}(4-\sqrt{h})}{20} = 0 \Rightarrow \right\} 4-\sqrt{h} = 0$	M1	3.4
	Deduces any of $0 < h < 16$, $0 \leq h < 16$, $0 < h \leq 16$, $0 \leq h \leq 16$, $h < 16$, $h \leq 16$ or all values up to 16	A1	2.2a
		(2)	
(c) Way 1	$\int \frac{1}{(4-\sqrt{h})} dh = \int \frac{1}{20} t^{0.25} dt$	B1	1.1b
	$-8\ln 4-\sqrt{h} - 2\sqrt{h} = \frac{1}{25} t^{1.25} \{+c\}$	M1	1.1b
		A1	1.1b
	$\{t=0, h=1 \Rightarrow\} -8\ln(4-1) - 2\sqrt{1} = \frac{1}{25}(0)^{1.25} + c$	M1	3.4
	$\Rightarrow c = -8\ln(3) - 2 \Rightarrow -8\ln 4-\sqrt{h} - 2\sqrt{h} = \frac{1}{25} t^{1.25} - 8\ln(3) - 2$ $\{h=12 \Rightarrow\} -8\ln 4-\sqrt{12} - 2\sqrt{12} = \frac{1}{25} t^{1.25} - 8\ln(3) - 2$	dM1	3.1a
	$t^{1.25} = 221.2795202... \Rightarrow t = \sqrt[1.25]{221.2795...} \text{ or } t = (221.2795...)^{0.8}$	M1	1.1b
	$t = 75.154... \Rightarrow t = 75.2 \text{ (years) (3 sf) or awrt 75.2 (years)}$	A1	1.1b
	Note: You can recover work for part (c) in part (b)	(7)	
(c) Way 2	$\int_1^{12} \frac{20}{(4-\sqrt{h})} dh = \int_0^T t^{0.25} dt$	B1	1.1b
	$\left[20(-8\ln 4-\sqrt{h} - 2\sqrt{h}) \right]_1^{12} = \left[\frac{4}{5} t^{1.25} \right]_0^T$	M1	1.1b
		A1	1.1b
	$20(-8\ln(4-\sqrt{12}) - 2\sqrt{12}) - 20(-8\ln(4-1) - 2\sqrt{1}) = \frac{4}{5} T^{1.25} - 0$	M1	3.4
		dM1	3.1a
	$T^{1.25} = 221.2795202... \Rightarrow T = \sqrt[1.25]{221.2795...} \text{ or } T = (221.2795...)^{0.8}$	M1	1.1b
	$T = 75.154... \Rightarrow T = 75.2 \text{ (years) (3 sf) or awrt 75.2 (years)}$	A1	1.1b
	Note: You can recover work for part (c) in part (b)	(7)	

(15 marks)

Question	Scheme	Marks	AOs
8	Any equation involving an exponential of the correct form. See notes	M1	3.1b
	$n = Ae^{kt}$ (where A and k are positive constants)	A1	1.1b
		(2)	
(2 marks)			
Notes:			

M1: Any equation of the correct form, involving n and an exponential in t .

So allow for example $n = e^{\pm t}$, $n = Ae^{\pm t}$, $n = Ae^{\pm kt}$ condoning $n = A + Be^{\pm t}$

Condone an intermediate form where n has not been made the subject.

E.g. Allow $\ln n = kt + c$ but also condone $\ln n = kt$

A1: E.g. $n = Ae^{kt}$, $n = e^{kt+c}$, $n = e^{kt} e^c$ There is no requirement to state that A and k are positive constants

Note that the two constants need to be different.

Mark the final answer so $n = e^{kt+c}$ followed by $n = e^{kt} + e^c$ o.e. $n = e^{kt} + A$ such as is M1 A0

.....
You may see solutions that don't include "e".

These are fine so you can include versions of $n = Ak^t$ using the same marking criteria as above

FYI $\frac{dn}{dt} = Ak^t \times \ln k = \ln k \times n$ so $\frac{dn}{dt} \propto n$

.....

Question	Scheme	Marks	AOs
14 (a)	Uses the model to state $\frac{dV}{dt} = -c$ (for positive constant c)	B1	3.1b
	Uses $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$ with their $\frac{dV}{dt} = -c$ and $\frac{dV}{dr} = 4\pi r^2$	M1	2.1
	$-c = 4\pi r^2 \times \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = -\frac{c}{4\pi r^2} = -\frac{k}{r^2}$ *	A1*	2.2a
		(3)	
(b)	$\frac{dr}{dt} = -\frac{k}{r^2} \Rightarrow \int r^2 dr = \int -k dt$ and integrates with one side "correct"	M1	2.1
	$\frac{r^3}{3} = -kt (+\alpha)$	A1	1.1b
	Uses $t = 0, r = 40 \Rightarrow \alpha = \dots$ $\alpha = \frac{64000}{3}$	M1	1.1b
	Uses $t = 5, r = 20$ & $\alpha = \dots \Rightarrow k = \dots$	M1	3.4
	$r^3 = 64000 - 11200t$ or exact equivalent	A1	3.3
		(5)	
(c)	Uses the equation of their model and proceeds to a limiting value for t E.g. " $64000 - 11200t$ " ... $0 \Rightarrow t \dots$	M1	3.4
	For times up to and including $\frac{40}{7}$ seconds	A1ft	3.5b
		(2)	
(10 marks)			
Notes:			

Question Number	Scheme	Marks
9(a)	$u = 4 - \sqrt{x} \Rightarrow x = (4 - u)^2 \Rightarrow \frac{dx}{du} = -2(4 - u)$ $\int \frac{dx}{4 - \sqrt{x}} = \int \frac{-2(4 - u)du}{u} = \int -\frac{8}{u} + 2du$ $= -8 \ln u + 2u \quad (+c)$ $= -8 \ln 4 - \sqrt{x} + 2(4 - \sqrt{x}) (+c) \quad \text{oe}$	M1A1 M1A1 dM1 A1 (6)
(b)	Height increases when $\frac{dh}{dt} = \frac{4 - \sqrt{h}}{20} > 0 \Rightarrow (0 <) h < 16$	M1A1 (2)
(c)	$\frac{dh}{dt} = \frac{4 - \sqrt{h}}{20} \Rightarrow \int \frac{dh}{4 - \sqrt{h}} = \int \frac{dt}{20}$ $\Rightarrow -8 \ln(4 - \sqrt{h}) + 2(4 - \sqrt{h}) = \frac{t}{20} + c$ Substitute $t=0, h=1$ $-8 \ln 3 + 6 = c$ $\Rightarrow -8 \ln(4 - \sqrt{h}) + 2(4 - \sqrt{h}) = \frac{t}{20} - 8 \ln 3 + 6 \quad \text{oe}$ Substitute $h=10$ into $\Rightarrow -8 \ln(4 - \sqrt{10}) + 2(4 - \sqrt{10}) = \frac{t}{20} - 8 \ln 3 + 6$ $\Rightarrow t = \text{awrt } 118 (\text{years})$	B1 M1A1 dM1 A1 ddM1 A1 (7) (15 marks)
Alt (c)	$\frac{dh}{dt} = \frac{4 - \sqrt{h}}{20} \Rightarrow \int \frac{dh}{4 - \sqrt{h}} = \int \frac{dt}{20}$ $\Rightarrow -8 \ln(4 - \sqrt{h}) + 2(4 - \sqrt{h}) = \frac{t}{20}$ $\Rightarrow \left[-8 \ln(4 - \sqrt{h}) + 2(4 - \sqrt{h}) \right]_{h=1}^{h=10} = \left[\frac{t}{20} \right]_{t=0}^T$ $\Rightarrow \left(-8 \ln(4 - \sqrt{10}) + 2(4 - \sqrt{10}) \right) - \left(-8 \ln(4 - \sqrt{1}) + 2(4 - \sqrt{1}) \right) = \frac{T}{20}$ $\Rightarrow T = \text{awrt } 118 (\text{years})$	B1 M1 A1 dM1, ddM1, A1 A1

Question Number	Scheme	Marks
6. (a)	$\frac{5-4x}{(2x-1)(x+1)} \equiv \frac{A}{(2x-1)} + \frac{B}{(x+1)} \quad \text{so} \quad 5-4x \equiv A(x+1) + B(2x-1)$	B1
	Let $x = -1$, $9 = B(-3) \Rightarrow B = -3$ Let $x = \frac{1}{2}$, $3 = A\left(\frac{3}{2}\right) \Rightarrow A = 2$	M1
	$A = 2$ and $B = -3$ or $\left\{ \frac{5-4x}{(2x-1)(x+1)} \equiv \frac{2}{(2x-1)} - \frac{3}{(x+1)} \right\}$	A1
(b)		[3]
(i), (ii)	$\int \frac{1}{y} dy = \int \frac{5-4x}{(2x-1)(x+1)} dx$ $= \int \frac{2}{(2x-1)} - \frac{3}{(x+1)} dx = C \ln(2x-1) + D \ln(x+1)$ $= \frac{1}{2} \ln(2x-1) - 3 \ln(x+1)$ $\ln y = \ln(2x-1) - 3 \ln(x+1) + c$	B1
		M1
		A1ft
Method 1 for (ii)	$\ln 4 = \ln(2(2)-1) - 3 \ln(2+1) + c \Rightarrow c = \{\ln 36\}$	A1
		M1
	$\ln y = \ln(2x-1) - 3 \ln(x+1) + \ln 36 \quad \text{so} \quad \ln y = \ln \left(\frac{36(2x-1)}{(x+1)^3} \right) \quad \text{So} \quad y = \frac{36(2x-1)}{(x+1)^3}$	M1 A1
		[7]
Method 2 for (ii)	Solution as Method 1 up to $\ln y = \ln(2x-1) - 3 \ln(x+1) + c$ so first four marks as before	B1M1A1A1
	Writes $y = \frac{A(2x-1)}{(x+1)^3}$ as general solution which would earn the 3 rd M1 mark.	M1
	Then may substitute to find their constant A , which would earn the 2 nd M1 mark.	M1
	Then A1 for $y = \frac{36(2x-1)}{(x+1)^3}$ as before.	A1
		[7]
		10

Question Number	Scheme	Marks
8 (a)	$\frac{d}{dy}(\ln \tan 2y) = \frac{1}{\tan 2y} \times 2 \sec^2 2y$ $= \frac{\cos 2y}{\sin 2y} \times \frac{2}{\cos^2 2y} = \frac{k}{\sin 2y \cos 2y}$ $= \frac{4}{2 \sin 2y \cos 2y} = \frac{4}{\sin 4y}$	M1A1 M1 A1* cso (4)
(b) Way 1	$\frac{dy}{dx} = 2 \cos x \sin 4y \Rightarrow \int \frac{dy}{\sin 4y} = \int 2 \cos x dx$ $\Rightarrow \frac{1}{4} \ln \tan 2y = 2 \sin x (+c)$	B1 M1A1
Finds limits first	Put $x = 0, y = \frac{\pi}{6} \Rightarrow \frac{1}{4} \ln \tan 2 \frac{\pi}{6} = 2 \sin 0 + c \Rightarrow c = \dots \left(\frac{1}{4} \ln \sqrt{3} \text{ or } \frac{1}{8} \ln 3 \right)$ Takes exponentials so $\tan 2y = e^{8 \sin x + c}$	M1 M1
Finds limits after removing lns	$\tan 2y = e^{8 \sin x + c}$ (so $\tan 2y = A e^{8 \sin x}$) Put $x = 0, y = \frac{\pi}{6}$, so $A =$ or $e^c =$	M1 (bM3 on open) M1 (bM2 on open)
	$\tan 2y = \sqrt{3} e^{8 \sin x}$	A1 (6) (10 marks)
(b) Way 2	$\frac{dy}{dx} = 2 \cos x \sin 4y \Rightarrow \int \frac{dy}{\sin 4y} = \int 2 \cos x dx$ $\Rightarrow -\frac{1}{4} \ln(\operatorname{cosec} 4y + \cot 4y) = 2 \sin x (+c)$ Sub $x = 0, y = \frac{\pi}{6}$ $\Rightarrow -\frac{1}{4} \ln(\operatorname{cosec} \frac{2\pi}{3} + \cot \frac{2\pi}{3}) = 2 \sin 0 + c \Rightarrow c = \dots \left(-\frac{1}{4} \ln \frac{1}{\sqrt{3}} \text{ or } \frac{1}{4} \ln \sqrt{3} \right)$ $-\frac{1}{4} \ln\left(\frac{1 + \cos 4y}{\sin 4y}\right) = \frac{1}{4} \ln(\tan 2y) = 2 \sin x + \frac{1}{4} \ln \sqrt{3} \quad \text{so } \tan 2y = \sqrt{3} e^{8 \sin x}$	B1 M1A1 M1 M1A1 (6)
(b) Way 3	Special case: Differentiates the answer. Marks available B0M1A0M1M1A1 $\tan 2y = A e^{B \sin x} \rightarrow (2) \sec^2 2y \frac{dy}{dx} = AB \cos x e^{B \sin x}$ $\frac{dy}{dx} = \frac{B \cos x \tan 2y \cos^2 2y}{(2)}$ $\frac{dy}{dx} = \frac{B \cos x 2 \sin 2y \cos 2y}{(4)} = \frac{B \cos x \sin 4y}{(4)}$ $B = 8 \text{ and } A = \sqrt{3}$	M1 M1 M1 A1 4/6

Question Number	Scheme	Marks
9(a)	$\frac{3x^2-4}{x^2(3x-2)} \equiv \frac{A}{x} + \frac{B}{x^2} + \frac{C}{3x-2}$ $\frac{2}{x^2}, \frac{-6}{3x-2} \quad (B=2, C=-6)$ $3x^2-4 \equiv Ax(3x-2)+B(3x-2)+Cx^2 \Rightarrow A=..$ $\frac{3}{x} \quad (A=3) \text{ is one of the fractions}$	B1, B1, M1 A1 [4]
(b)	$\int \frac{1}{y} dy = \int \frac{3x^2-4}{x^2(3x-2)} dx$ $\ln y = \int \frac{A}{x} + \frac{B}{x^2} + \frac{C}{3x-2} dx$ $= A \ln x - \frac{B}{x} + \frac{C}{3} \ln(3x-2) \quad (+k)$ $y = e^{A \ln x - \frac{B}{x} + \frac{C}{3} \ln(3x-2) + D} \quad \text{or} \quad y = D e^{A \ln x - \frac{B}{x} + \frac{C}{3} \ln(3x-2)}$ $y = K x^3 (3x-2)^{-2} e^{-\frac{2}{x}} \quad \text{or} \quad \frac{K x^3 e^{-\frac{2}{x}}}{(3x-2)^2} \quad \text{or} \quad \frac{e^k x^3 e^{-\frac{2}{x}}}{(3x-2)^2} \quad \text{oe}$	B1 M1 M1A1ft M1 A1cso [6] (10 marks)

Question Number	Scheme	Marks	
13(a)	$V = \frac{1}{3}\pi h^2(30 - h) = 10\pi h^2 - \frac{1}{3}\pi h^3 \Rightarrow \frac{dV}{dh} = 20\pi h - \pi h^2$ or $V = \frac{1}{3}\pi h^2(30 - h) \Rightarrow \frac{dV}{dh} = \frac{2}{3}\pi h(30 - h) - \frac{1}{3}\pi h^2$	M1A1	
	M1: Attempts $\frac{dV}{dh}$ either by multiplying out and differentiating each term to give a derivative of the form $\alpha h - \beta h^2$ or by the product rule to give a derivative of the form $\alpha h(30 - h) \pm \beta h^2$. A1: Any correct (possibly un-simplified) form for $\frac{dV}{dh}$		
	Uses $\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt} \Rightarrow -\frac{1}{10}V = (20\pi h - \pi h^2) \times \frac{dh}{dt}$	M1	
	Uses a correct form of the chain rule, e.g. $\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$ or uses $\frac{dh}{dV} \times \frac{dV}{dt}$ with their $\frac{dV}{dh}$ and $\frac{dV}{dt} = -\frac{1}{10}V$.		
	$\Rightarrow -\frac{1}{10} \times \frac{1}{3}\pi h^2(30 - h) = \pi h(20 - h) \times \frac{dh}{dt} \left(\Rightarrow \frac{dh}{dt} = \dots \right)$	M1	
	Substitutes $V = \frac{1}{3}\pi h^2(30 - h)$ and rearranges to obtain $\frac{dh}{dt}$ in terms of h		
	$\frac{dh}{dt} = -\frac{h(30 - h)}{30(20 - h)} *$	This is a given answer. There must have been intermediate lines and correct factorisation and no errors and " $\frac{dh}{dt} =$ " must be seen at some point.	A1*
		(5)	
(b)	$\frac{30(20 - h)}{h(30 - h)} \equiv \frac{A}{h} + \frac{B}{30 - h}$	Correct form for the partial fractions	B1
	$30(20 - h) \equiv A(30 - h) + Bh$ $h = 30 \Rightarrow 30B = -300 \Rightarrow B = -10 \text{ and } h = 0 \Rightarrow 30A = 600 \Rightarrow A = 20$ Attempts to get both constants by a correct method e.g. substituting, comparing coefficients, cover up rule		M1
	$\frac{30(20 - h)}{h(30 - h)} \equiv \frac{20}{h} - \frac{10}{30 - h}$	Correct partial fractions (or states " A " = 20, " B " = -10)	A1
		(3)	

(c) Way 1		
	$\frac{dh}{dt} = -\frac{h(30-h)}{30(20-h)} \Rightarrow \int \frac{30(20-h)}{h(30-h)} dh = -1 \int dt$ A correct statement which may be implied by subsequent work. Condone the omission of “dh” and “dt” provided the intention is clear but the minus sign must be present on one side or the other.	B1
	$20 \ln h + 10 \ln(30 - h)$ M1: Integrates their partial fractions to obtain $\pm P \ln h \pm Q \ln(30 - h)$ A1: Correct integration for their partial fractions of the form $\frac{A}{h} + \frac{B}{30 - h}$ following through their “A” and “B”.	M1A1ft
	$t = 0, h = 10 \Rightarrow c = 20 \ln 10 + 10 \ln 20$	M1
	$h = 5 \Rightarrow t = 20 \ln 10 + 10 \ln 20 - 10 \ln 25 - 20 \ln 5$ Substitutes $h = 5$ and uses their value of c to find a value for t .	ddM1
	$t = 11.63$ (secs)	A1cso
		(6)
		(14 marks)
(c) Way 2		
	$\frac{dh}{dt} = -\frac{h(30-h)}{30(20-h)} \Rightarrow \int \frac{30(20-h)}{h(30-h)} dh = -1 \int dt$ A correct statement which may be implied by subsequent work. Condone the omission of “dh” and “dt” provided the intention is clear but the minus sign must be present on one side or the other.	B1
	$20 \ln h + 10 \ln(30 - h)$ M1: Integrates their partial fractions to obtain $\pm P \ln h \pm Q \ln(30 - h)$ A1: Correct integration for their partial fractions of the form $\frac{A}{h} + \frac{B}{30 - h}$ following through their “A” and “B”.	M1A1ft
	$(t =) [20 \ln h + 10 \ln(30 - h)]_5^{10}$ or $(t =) [20 \ln h + 10 \ln(30 - h)]_{10}^5$	M1
	$(t =) [20 \ln 10 + 10 \ln 20] - [20 \ln 5 + 10 \ln 25]$	ddM1
	$t = 11.63$	A1cso
		(6)

Qu	Scheme		Marks
12. (a)	Way 1: Uses $x = kt$ or $t = cx$ and $x = 1.5$ when $t = 2$ so $k =$ or $c =$ $t = \frac{4}{3}x$	Way 2: Uses $x = kt + c$ with $x = 0, t = 0$ and with $x = 1.5$ when $t = 2$ so $k =$ $t = \frac{4}{3}x$	M1
	(b) $t = 4$		A1 (2)
	(c) $\frac{dx}{dt} = \frac{\lambda}{(2x+1)}$ so separate variables to give $\int (2x+1) dx = \int \lambda dt$ $x^2 + x = \lambda t(+c')$ or $\frac{(2x+1)^2}{4} = \lambda t(+c)$ so $t =$		M1
	(When $t = 0, x = 0$ so $c = 1/4$ or $c' = 0$) so $t = \frac{x^2 + x}{\lambda}$		M1
	(d) Uses $x = 1.5$ when $t = 2$ to give $\lambda = \frac{15}{8}$		A1 (3)
	(e) $t = \frac{x^2 + x}{\lambda} = \frac{12}{\lambda} = 6.4$ hours later <u>so</u> <u>10.24pm or 22.24</u>		B1 (1)
			M1 <u>A1</u> (2) (9 marks)

Mark (a) and (b) together

(a)

M1: Uses correct $x = kt$ or $t = cx$ and $x = 1.5$ when $t = 2$ to find their constant (may not be k or c)

This may be the result of a differential equation $\frac{dx}{dt} = k$

A1: $t = \frac{4}{3}x$ oe such as $t = \frac{x}{0.75}$ or even $t = \frac{x}{3/4}$ Just this with no working is M1 A1

(b)

B1: $t = 4$

Mark (c),(d) and (e) together

(c)

M1: Correct separation but condone missing integral signs

M1: Correct form for both integrals- may not find c or even include a c

A1: Obtains a correct answer for t in terms of x and λ by using either $x = 0, t = 0 \Rightarrow t = \frac{x^2 + x}{\lambda}$ or

$$t = \frac{(2x+1)^2 - 1}{4\lambda} \text{ oe. Alternatively uses } x = 1.5, t = 2 \Rightarrow t = \frac{4x^2 + 4x + 8\lambda - 15}{4\lambda} \text{ oe}$$

Condone correct responses where 'c' seems to have been either cancelled out or ignored

(d)

B1: $\lambda = \frac{15}{8}$ or decimal i.e. 1.875

(e)

M1: Substitutes $x = 3$ into their expression for t . Implied by $t = \frac{12}{\lambda}$

A1: 10.24pm or 22:24 only

Question Number	Scheme	Marks
12 (a)	States or uses $\frac{dV}{dt} = 0.4\pi - 0.2\pi\sqrt{h}$	B1
	States or uses $V = 4\pi h \Rightarrow \frac{dV}{dh} = 4\pi$	M1A1
	Uses $\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt} \Rightarrow 0.4\pi - 0.2\pi\sqrt{h} = 4\pi \times \frac{dh}{dt}$	M1
	$20 \frac{dh}{dt} = 2 - \sqrt{h}$	A1*
		(5)
(b)	Separates the variables $\int 20 \frac{dh}{2 - \sqrt{h}} = \int 1 dt$	M1
	$(t =) \int_{0.16}^{2.25} \frac{20}{2 - \sqrt{h}} dh$	A1*
		(2)
(c)	$h = (2 - x)^2 \Rightarrow dh = -2(2 - x)dx$	B1
	$T = \int \frac{20}{2 - \sqrt{h}} dh = \int \frac{20}{2 - (2 - x)} \times -2(2 - x) dx$	M1
	$= \int -\frac{80}{x} + 40 dx$ (No need for limits here)	dM1A1
	$T = \int_{0.16}^{2.25} \frac{20}{2 - \sqrt{h}} dh = \int_{1.6}^{0.5} -\frac{80}{x} + 40 dx = [-80 \ln x + 40x]_{1.6}^{0.5}$	ddM1
	$= [-80 \ln 0.5 + 20] - [-80 \ln 1.6 + 64] = 80 \ln 3.2 - 44 = 49 \text{ (minutes)}$	dddM1A1
		(7)
		(14 marks)

(a)

B1: States or uses $\frac{dV}{dt} = 0.4\pi - 0.2\pi\sqrt{h}$. This may be embedded within the chain rule but must be

identifiable as $\frac{dV}{dt}$

M1: Attempts $\frac{dV}{dh}$ from an equation for the volume of a cylinder. Accept $V = c \times \pi h \rightarrow \frac{dV}{dh} = c\pi$

A1: $\frac{dV}{dh} = 4\pi$ This may be embedded within the chain rule

M1: Uses a correct form of the chain rule: Eg $\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$ with their $\frac{dV}{dh}$ and $\frac{dh}{dt}$

Also accept forms such as $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$

A1*: $20 \frac{dh}{dt} = 2 - \sqrt{h}$ This is a given answer so no errors must be seen e.g. bracketing errors.

Question Number	Scheme	Marks
10 (a)	$t(t-4) = 0 \Rightarrow t = 4$ Hence $x = \frac{20 \times 4}{2 \times 4 + 1} = \frac{80}{9}$	M1A1
		(2)
(b)	$x = \frac{20t}{2t+1} \Rightarrow \frac{dx}{dt} = \frac{20(2t+1) - 20t \times 2}{(2t+1)^2} = \left(\frac{20}{(2t+1)^2} \right)$	M1A1
	$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{(2t-4)}{20} \cdot \frac{(2t+1)^2}{(2t+1)^2} = \frac{(t-2)(2t+1)^2}{10}$	M1A1,A1
		(5)
Mark c(i) and (ii) together:		
(c)(i)	$x = \frac{20t}{2t+1} \Rightarrow 2tx + x = 20t \Rightarrow t(20-2x) = x \Rightarrow t = \frac{x}{20-2x}$ or $\frac{-x}{2x-20}$	M1A1
		(2)
(ii)	Sub $t = \frac{x}{20-2x}$ into $y = t(t-4) \Rightarrow y = \frac{x}{20-2x} \left(\frac{x}{20-2x} - 4 \right)$	M1
	$\Rightarrow y = \frac{x}{20-2x} \times \left(\frac{x}{20-2x} - \frac{4(20-2x)}{20-2x} \right)$	dM1
	$\Rightarrow y = \frac{x}{20-2x} \times \left(\frac{9x-80}{20-2x} \right)$	
	$\Rightarrow y = \frac{x(9x-80)}{(20-2x)^2}$, oe $0 < x < 10$ or $k = 10$	A1, B1
		(4)
		(13 marks)

(a)

M1: Attempts to find x when $t = 4$ A1: $\frac{80}{9}$ (Not 8.88... but isw if $\frac{80}{9}$ is seen)(Ignore any attempts to find x when $t = 0$)

(b)

M1: Attempts to apply the quotient rule on $\frac{20t}{2t+1}$ with $u = 20t, v = 2t+1$ Alternatively applies the product rule on $20t(2t+1)^{-1}$ OR writes $\frac{20t}{2t+1}$ as $A - \frac{B}{2t+1}$ and uses the chain ruleA1: $\frac{dx}{dt} = \frac{20(2t+1) - 20t \times 2}{(2t+1)^2}$ or $\frac{dx}{dt} = 20(2t+1)^{-1} + 20t \times -2(2t+1)^{-2}$

Question Number	Scheme		Notes	Marks
7.	$\frac{dh}{dt} = k\sqrt{h-9}$, $9 < h \leq 200$; $h = 130$, $\frac{dh}{dt} = -1.1$			
(a)	$-1.1 = k\sqrt{(130-9)} \Rightarrow k = \dots$	Substitutes $h = 130$ and either $\frac{dh}{dt} = -1.1$ or $\frac{dh}{dt} = 1.1$ into the printed equation and rearranges to give $k = \dots$		M1
	so, $k = -\frac{1}{10}$ or -0.1	$k = -\frac{1}{10}$ or -0.1		A1
				[2]
(b) Way 1	$\int \frac{dh}{\sqrt{h-9}} = \int k dt$	Separates the variables correctly. dh and dt should not be in the wrong positions, although this mark can be implied by later working. Ignore the integral signs.		B1
	$\int (h-9)^{-\frac{1}{2}} dh = \int k dt$			
	$\frac{(h-9)^{\frac{1}{2}}}{(\frac{1}{2})} = kt (+c)$	Integrates $\frac{\pm \lambda}{\sqrt{h-9}}$ to give $\pm m\sqrt{h-9}$; $l, m^1 0$		M1
		$\frac{(h-9)^{\frac{1}{2}}}{(\frac{1}{2})} = kt$ or $\frac{(h-9)^{\frac{1}{2}}}{(\frac{1}{2})} = (\text{their } k)t$, with/without $+c$, or equivalent, which can be un-simplified or simplified.		A1
	$\{t = 0, h = 200 \Rightarrow\} 2\sqrt{(200-9)} = k(0) + c$	Some evidence of applying both $t = 0$ and $h = 200$ to changed equation containing a constant of integration, e.g. c or A		M1
	$\Rightarrow c = 2\sqrt{191} \Rightarrow 2(h-9)^{\frac{1}{2}} = -0.1t + 2\sqrt{191}$ $\{h = 50 \Rightarrow\} 2\sqrt{(50-9)} = -0.1t + 2\sqrt{191}$ $t = \dots$	dependent on the previous M mark Applies $h = 50$ and their value of c to their changed equation and rearranges to find the value of $t = \dots$		dM1
	$t = 20\sqrt{191} - 20\sqrt{41}$ or $t = 148.3430145\dots = 148$ (minutes) (nearest minute)	$t = 20\sqrt{191} - 20\sqrt{41}$ isw or awrt 148		A1 cso
				[6]
	$\int_{200}^{50} \frac{dh}{\sqrt{h-9}} = \int_0^T k dt$	Separates the variables correctly. dh and dt should not be in the wrong positions, although this mark can be implied by later working. Integral signs and limits not necessary.		B1
	$\int_{200}^{50} (h-9)^{-\frac{1}{2}} dh = \int_0^T k dt$			
(b) Way 2	$\left[\frac{(h-9)^{\frac{1}{2}}}{(\frac{1}{2})} \right]_{200}^{50} = [kt]_0^T$	Integrates $\frac{\pm \lambda}{\sqrt{h-9}}$ to give $\pm m\sqrt{h-9}$; $l, m^1 0$		M1
		$\frac{(h-9)^{\frac{1}{2}}}{(\frac{1}{2})} = kt$ or $\frac{(h-9)^{\frac{1}{2}}}{(\frac{1}{2})} = (\text{their } k)t$, with/without limits, or equivalent, which can be un-simplified or simplified.		A1
	$2\sqrt{41} - 2\sqrt{191} = kt$ or kT	Attempts to apply limits of $h = 200$, $h = 50$ and (can be implied) $t = 0$ to their changed equation		M1
	$t = \frac{2\sqrt{41} - 2\sqrt{191}}{-0.1}$	dependent on the previous M mark Then rearranges to find the value of $t = \dots$		dM1
	$t = 20\sqrt{191} - 20\sqrt{41}$ or $t = 148.3430145\dots = 148$ (minutes) (nearest minute)	$t = 20\sqrt{191} - 20\sqrt{41}$ or awrt 148 or 2 hours and awrt 28 minutes		A1 cso
				[6]
				8

Question Number	Scheme	Marks
7. (a)	$\frac{2}{P(P-2)} = \frac{A}{P} + \frac{B}{(P-2)}$	
	$2 \equiv A(P-2) + BP$	Can be implied. M1
	$A = -1, B = 1$	Either one. A1
	giving $\frac{1}{(P-2)} - \frac{1}{P}$	See notes. cao, aef A1
		[3]
	(b) $\frac{dP}{dt} = \frac{1}{2}P(P-2)\cos 2t$	
	$\int \frac{2}{P(P-2)} dP = \int \cos 2t dt$	can be implied by later working B1 oe
	$\ln(P-2) - \ln P = \frac{1}{2}\sin 2t (+c)$	$\pm \lambda \ln(P-2) \pm \mu \ln P,$ $\lambda \neq 0, \mu \neq 0$ M1
	$\{t=0, P=3 \Rightarrow\} \ln 1 - \ln 3 = 0 + c \quad \{\Rightarrow c = -\ln 3 \text{ or } \ln(\frac{1}{3})\}$	$\ln(P-2) - \ln P = \frac{1}{2}\sin 2t$ A1 See notes M1
	$\ln(P-2) - \ln P = \frac{1}{2}\sin 2t - \ln 3$ $\ln\left(\frac{3(P-2)}{P}\right) = \frac{1}{2}\sin 2t$	
(c)	$\frac{3(P-2)}{P} = e^{\frac{1}{2}\sin 2t}$	Starting from an equation of the form $\pm \lambda \ln(P-\beta) \pm \mu \ln P = \pm K \sin \delta t + c,$ $\lambda, \mu, \beta, K, \delta \neq 0$, applies a fully correct method to eliminate their logarithms. Must have a constant of integration that need not be evaluated (see note) M1
	$3(P-2) = Pe^{\frac{1}{2}\sin 2t} \Rightarrow 3P - 6 = Pe^{\frac{1}{2}\sin 2t}$	A complete method of rearranging to make P the subject.
	gives $3P - Pe^{\frac{1}{2}\sin 2t} = 6 \Rightarrow P(3 - e^{\frac{1}{2}\sin 2t}) = 6$	Must have a constant of integration that need not be evaluated (see note) dM1
	$P = \frac{6}{(3 - e^{\frac{1}{2}\sin 2t})} *$	Correct proof. A1 * cso
		[7]
	$\{\text{population} = 4000 \Rightarrow\} P = 4$	States $P = 4$ or applies $P = 4$ M1
	$\frac{1}{2}\sin 2t = \ln\left(\frac{3(4-2)}{4}\right) \left\{ = \ln\left(\frac{3}{2}\right) \right\}$	Obtains $\pm \lambda \sin 2t = \ln k$ or $\pm \lambda \sin t = \ln k,$ $\lambda \neq 0, k > 0$ where λ and k are numerical values and λ can be 1 M1
	$t = 0.4728700467...$	anything that rounds to 0.473 Do not apply isw here A1
		[3]
		13

Question Number	Scheme		Marks
7. (b)	Method 2 for Q7(b)		
	$\ln(P-2) - \ln P = \frac{1}{2} \sin 2t \quad (+c)$	As before for...	B1M1A1
	$\ln\left(\frac{P-2}{P}\right) = \frac{1}{2} \sin 2t + c$		
	$\frac{(P-2)}{P} = e^{\frac{1}{2} \sin 2t + c}$ or $\frac{(P-2)}{P} = Ae^{\frac{1}{2} \sin 2t}$	Starting from an equation of the form $\pm \lambda \ln(P-\beta) \pm \mu \ln P = \pm K \sin \delta t + c$, $\lambda, \mu, \beta, K, \delta \neq 0$, applies a fully correct method to eliminate their logarithms. Must have a constant of integration that need not be evaluated (see note)	3rd M1
	$(P-2) = APe^{\frac{1}{2} \sin 2t} \Rightarrow P - APe^{\frac{1}{2} \sin 2t} = 2$	A complete method of rearranging to make P the subject. Condone sign slips or constant errors. Must have a constant of integration that need not be evaluated (see note)	4th dM1
	$\Rightarrow P(1 - Ae^{\frac{1}{2} \sin 2t}) = 2 \Rightarrow P = \frac{2}{(1 - Ae^{\frac{1}{2} \sin 2t})}$	See notes (Allocate this mark as the 2nd M1 mark on ePEN).	2nd M1
	$\left\{ \Rightarrow 3 = \frac{2}{(1-A)} \Rightarrow A = \frac{1}{3} \right\}$		
	$\Rightarrow P = \frac{2}{\left(1 - \frac{1}{3}e^{\frac{1}{2} \sin 2t}\right)} \Rightarrow P = \frac{6}{(3 - e^{\frac{1}{2} \sin 2t})}^*$	Correct proof.	A1 * cso
Question 7 Notes			
7. (a)	M1	Forming a correct identity. For example, $2 \equiv A(P-2) + BP$ from $\frac{2}{P(P-2)} = \frac{A}{P} + \frac{B}{(P-2)}$	
	Note	A and B are not referred to in question.	
	A1	Either one of $A = -1$ or $B = 1$.	
	A1	$\frac{1}{(P-2)} - \frac{1}{P}$ or any equivalent form. This answer cannot be recovered from part (b).	
	Note	M1A1A1 can also be given for a candidate who finds both $A = -1$ and $B = 1$ and $\frac{A}{P} + \frac{B}{(P-2)}$ is seen in their working.	
	Note	Candidates can use ‘cover-up’ rule to write down $\frac{1}{(P-2)} - \frac{1}{P}$, so as to gain all three marks.	
	Note	Equating coefficients from $2 \equiv A(P-2) + BP$ gives $A+B=2, -2A=2 \Rightarrow A=-1, B=1$	

Question Number	Scheme	Marks												
8.	$\frac{dx}{dt} = k(M - x)$, where M is a constant													
(a)	$\frac{dx}{dt}$ is the <u>rate of increase</u> of the <u>mass of waste</u> products.	Any one correct explanation. B1												
	M is the <u>total mass</u> of <u>unburned fuel</u> and <u>waste fuel</u> (or the <u>initial mass</u> of <u>unburned fuel</u>)	Both explanations are correct. B1												
(b)	$\int \frac{1}{M-x} dx = \int k dt \quad \text{or} \quad \int \frac{1}{k(M-x)} dx = \int dt$ $-\ln(M-x) = kt \{+c\} \quad \text{or} \quad -\frac{1}{k} \ln(M-x) = t \{+c\}$ $\{t = 0, x = 0 \Rightarrow\} -\ln(M-0) = k(0) + c$ $c = -\ln M \Rightarrow -\ln(M-x) = kt - \ln M$ <table><tr><td><i>then either...</i></td><td><i>or...</i></td></tr><tr><td>$-kt = \ln(M-x) - \ln M$</td><td>$kt = \ln M - \ln(M-x)$</td></tr><tr><td>$-kt = \ln\left(\frac{M-x}{M}\right)$</td><td>$kt = \ln\left(\frac{M}{M-x}\right)$</td></tr><tr><td>$e^{-kt} = \frac{M-x}{M}$</td><td>$e^{kt} = \frac{M}{M-x}$</td></tr><tr><td>$Me^{-kt} = M-x$</td><td>$(M-x)e^{kt} = M$</td></tr><tr><td colspan="2">leading to $x = M - Me^{-kt}$ or $x = M(1 - e^{-kt})$ oe</td></tr></table>	<i>then either...</i>	<i>or...</i>	$-kt = \ln(M-x) - \ln M$	$kt = \ln M - \ln(M-x)$	$-kt = \ln\left(\frac{M-x}{M}\right)$	$kt = \ln\left(\frac{M}{M-x}\right)$	$e^{-kt} = \frac{M-x}{M}$	$e^{kt} = \frac{M}{M-x}$	$Me^{-kt} = M-x$	$(M-x)e^{kt} = M$	leading to $x = M - Me^{-kt}$ or $x = M(1 - e^{-kt})$ oe		See notes M1 A1 See notes M1
<i>then either...</i>	<i>or...</i>													
$-kt = \ln(M-x) - \ln M$	$kt = \ln M - \ln(M-x)$													
$-kt = \ln\left(\frac{M-x}{M}\right)$	$kt = \ln\left(\frac{M}{M-x}\right)$													
$e^{-kt} = \frac{M-x}{M}$	$e^{kt} = \frac{M}{M-x}$													
$Me^{-kt} = M-x$	$(M-x)e^{kt} = M$													
leading to $x = M - Me^{-kt}$ or $x = M(1 - e^{-kt})$ oe														